Filter Summary Report: CG,TIA,Automated,NO,L,simple,Z3,Z5

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Contents

1 Examined $H(z)$ for CG TIA Automated NO L simple Z3 Z5: $\frac{Z_3 Z_5 g_m - Z_3}{2 Z_3 g_m + Z_5 g_m + 1}$

$$H(z) = \frac{Z_3 Z_5 g_m - Z_3}{2 Z_3 g_m + Z_5 g_m + 1}$$

2 AP

3 BP

4 BS

5 GE

6 HP

7 LP

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(\infty,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \frac{R_5}{C_5 R_5 s + 1},\ \infty\right)$

$$H(s) = \frac{-C_5 R_5 s + R_5 g_m - 1}{C_3 C_5 R_5 s^2 + 2 g_m + s \left(C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m\right)}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{C_3} \sqrt{C_5} \sqrt{R_5} \sqrt{g_m}}{C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m}$
wo: $\frac{\sqrt{2} \sqrt{g_m}}{\sqrt{C_3} \sqrt{C_5} \sqrt{R_5}}$
bandwidth: $\frac{C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m}{C_3 C_5 R_5}$
K-LP: $\frac{R_5 g_m - 1}{2 g_m}$
K-HP: 0
K-BP: $-\frac{C_5 R_5}{C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\infty,\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_5 R_3 s + R_3 g_m}{C_3 C_5 R_3 s^2 + g_m + s \left(C_3 R_3 g_m + 2 C_5 R_3 g_m + C_5\right)}$$

Parameters:

Q: $\frac{\sqrt{C_3} \sqrt{C_5} \sqrt{R_3} \sqrt{g_m}}{C_3 R_3 g_m + 2 C_5 R_3 g_m + C_5}$
wo: $\frac{\sqrt{g_m}}{\sqrt{C_3} \sqrt{C_5} \sqrt{R_3}}$
bandwidth: $\frac{C_3 R_3 g_m + 2 C_5 R_3 g_m + C_5}{C_3 C_5 R_3}$
K-LP: R_3
K-HP: 0
K-BP: $-\frac{C_5 R_3}{C_3 R_3 g_m + 2 C_5 R_3 g_m + C_5}$
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\infty,\ \infty,\ \frac{R_3}{C_3R_3s+1},\ \infty,\ \frac{R_5}{C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{-C_5R_3R_5s + R_3R_5g_m - R_3}{C_3C_5R_3R_5s^2 + 2R_3g_m + R_5g_m + s\left(C_3R_3R_5g_m + C_3R_3 + 2C_5R_3R_5g_m + C_5R_5\right) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{R_5}\sqrt{2R_3g_m+R_5g_m+1}}{C_3R_3R_5g_m+C_3R_3+2C_5R_3R_5g_m+C_5R_5}$
wo: $\frac{\sqrt{2R_3g_m+R_5g_m+1}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{R_5}}$
bandwidth: $\frac{C_3R_3R_5g_m+C_3R_3+2C_5R_3R_5g_m+C_5R_5}{C_3C_5R_3R_5}$
K-LP: $\frac{R_3R_5g_m-R_3}{2R_3g_m+R_5g_m+1}$
K-HP: 0
K-BP: $-\frac{C_5R_3R_5}{C_3R_3R_5g_m+C_3R_3+2C_5R_3R_5g_m+C_5R_5}$
Qz: None
Wz: None

8.4 X-INVALID-NUMER-4 $Z(s) = \left(\infty,\ \infty,\ \frac{R_3}{C_3R_3s+1},\ \infty,\ R_5 + \frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{R_3g_m + s\left(C_5R_3R_5g_m - C_5R_3\right)}{g_m + s^2\left(C_3C_5R_3R_5g_m + C_3C_5R_3\right) + s\left(C_3R_3g_m + 2C_5R_3g_m + C_5R_5g_m + C_5\right)}$$

Parameters:

Q: $\frac{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}R_5g_m\sqrt{\frac{g_m}{R_5g_m+1}}+\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{\frac{g_m}{R_5g_m+1}}}{C_3R_3g_m+2C_5R_3g_m+C_5R_5g_m+C_5}$
wo: $\sqrt{\frac{g_m}{C_3C_5R_3R_5g_m+C_3C_5R_3}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{C_3C_5R_3R_5g_m+C_3C_5R_3}}\left(C_3R_3g_m+2C_5R_3g_m+C_5R_5g_m+C_5\right)}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}R_5g_m\sqrt{\frac{g_m}{R_5g_m+1}}+\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{\frac{g_m}{R_5g_m+1}}}$
K-LP: R_3
K-HP: 0
K-BP: $\frac{C_5R_3R_5g_m-C_5R_3}{C_3R_3g_m+2C_5R_3g_m+C_5R_5g_m+C_5}$
Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = \left(\infty,\ \infty,\ R_3,\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{R_3R_5g_m - R_3}{2R_3g_m + R_5g_m + 1}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(\infty,\ \infty,\ R_3,\ \infty,\ \frac{1}{C_5s},\ \infty\right)$

$$H(s) = \frac{-C_5R_3s + R_3g_m}{g_m + s\left(2C_5R_3g_m + C_5\right)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(\infty,\ \infty,\ R_3,\ \infty,\ \frac{R_5}{C_5R_5s+1},\ \infty\right)$

$$H(s) = \frac{-C_5R_3R_5s + R_3R_5g_m - R_3}{2R_3g_m + R_5g_m + s\left(2C_5R_3R_5g_m + C_5R_5\right) + 1}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(\infty,\ \infty,\ R_3,\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{R_3 g_m + s \left(C_5 R_3 R_5 g_m - C_5 R_3\right)}{g_m + s \left(2 C_5 R_3 g_m + C_5 R_5 g_m + C_5\right)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(\infty,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{R_5 g_m - 1}{2 g_m + s \left(C_3 R_5 g_m + C_3\right)}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(\infty,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_5 s + g_m}{C_3 C_5 s^2 + s \left(C_3 g_m + 2 C_5 g_m\right)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(\infty,\ \infty,\ \frac{1}{C_3 s},\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{g_m + s \left(C_5 R_5 g_m - C_5\right)}{s^2 \left(C_3 C_5 R_5 g_m + C_3 C_5\right) + s \left(C_3 g_m + 2 C_5 g_m\right)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(\infty,\ \infty,\ \frac{R_3}{C_3 R_3 s + 1},\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{R_3 R_5 g_m - R_3}{2 R_3 g_m + R_5 g_m + s \left(C_3 R_3 R_5 g_m + C_3 R_3\right) + 1}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(\infty,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ R_5,\ \infty\right)$

$$H(s) = \frac{R_5 g_m + s \left(C_3 R_3 R_5 g_m - C_3 R_3\right) - 1}{2 g_m + s \left(2 C_3 R_3 g_m + C_3 R_5 g_m + C_3\right)}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(\infty,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{-C_3 C_5 R_3 s^2 + g_m + s \left(C_3 R_3 g_m - C_5\right)}{s^2 \left(2 C_3 C_5 R_3 g_m + C_3 C_5\right) + s \left(C_3 g_m + 2 C_5 g_m\right)}$$

9.11 X-INVALID-ORDER-11 $Z(s) = \left(\infty,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ R_5 + \frac{1}{C_5 s},\ \infty\right)$

$$H(s) = \frac{g_m + s^2 \left(C_3 C_5 R_3 R_5 g_m - C_3 C_5 R_3\right) + s \left(C_3 R_3 g_m + C_5 R_5 g_m - C_5\right)}{s^2 \left(2 C_3 C_5 R_3 g_m + C_3 C_5 R_5 g_m + C_3 C_5\right) + s \left(C_3 g_m + 2 C_5 g_m\right)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\infty,\ \infty,\ R_3 + \frac{1}{C_3 s},\ \infty,\ \frac{R_5}{C_5 R_5 s + 1},\ \infty\right)$

$$H(s) = \frac{-C_3 C_5 R_3 R_5 s^2 + R_5 g_m + s\left(C_3 R_3 R_5 g_m - C_3 R_3 - C_5 R_5\right) - 1}{2 g_m + s^2\left(2 C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_5\right) + s\left(2 C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m\right)}$$

Parameters:

Q: $\frac{2\sqrt{2}\sqrt{C_3}\sqrt{C_5} R_3 \sqrt{R_5} g_m \sqrt{\frac{g_m}{2 R_3 g_m + 1}} + \sqrt{2}\sqrt{C_3}\sqrt{C_5}\sqrt{R_5}\sqrt{\frac{g_m}{2 R_3 g_m + 1}}}{2 C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m}$
wo: $\sqrt{2}\sqrt{\frac{g_m}{2 C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_5}}$
bandwidth: $\frac{\sqrt{2}\sqrt{\frac{g_m}{2 C_3 C_5 R_3 R_5 g_m + C_3 C_5 R_5}}\left(2 C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m\right)}{2\sqrt{2}\sqrt{C_3}\sqrt{C_5} R_3 \sqrt{R_5} g_m \sqrt{\frac{g_m}{2 R_3 g_m + 1}} + \sqrt{2}\sqrt{C_3}\sqrt{C_5}\sqrt{R_5}\sqrt{\frac{g_m}{2 R_3 g_m + 1}}}$
K-LP: $\frac{R_5 g_m - 1}{2 g_m}$
K-HP: $-\frac{R_3}{2 R_3 g_m + 1}$
K-BP: $\frac{C_3 R_3 R_5 g_m - C_3 R_3 - C_5 R_5}{2 C_3 R_3 g_m + C_3 R_5 g_m + C_3 + 2 C_5 R_5 g_m}$
Qz: None
Wz: $\frac{\sqrt{-R_5 g_m + 1}}{\sqrt{C_3}\sqrt{C_5}\sqrt{R_3}\sqrt{R_5}}$

11 X-PolynomialError